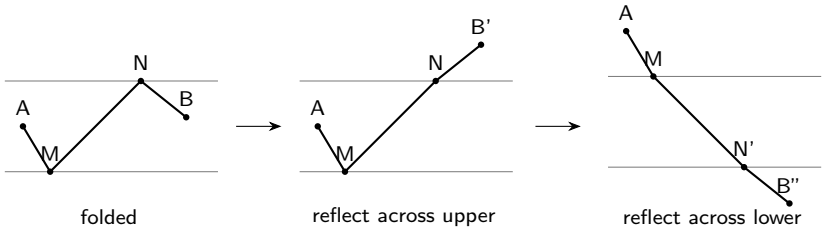
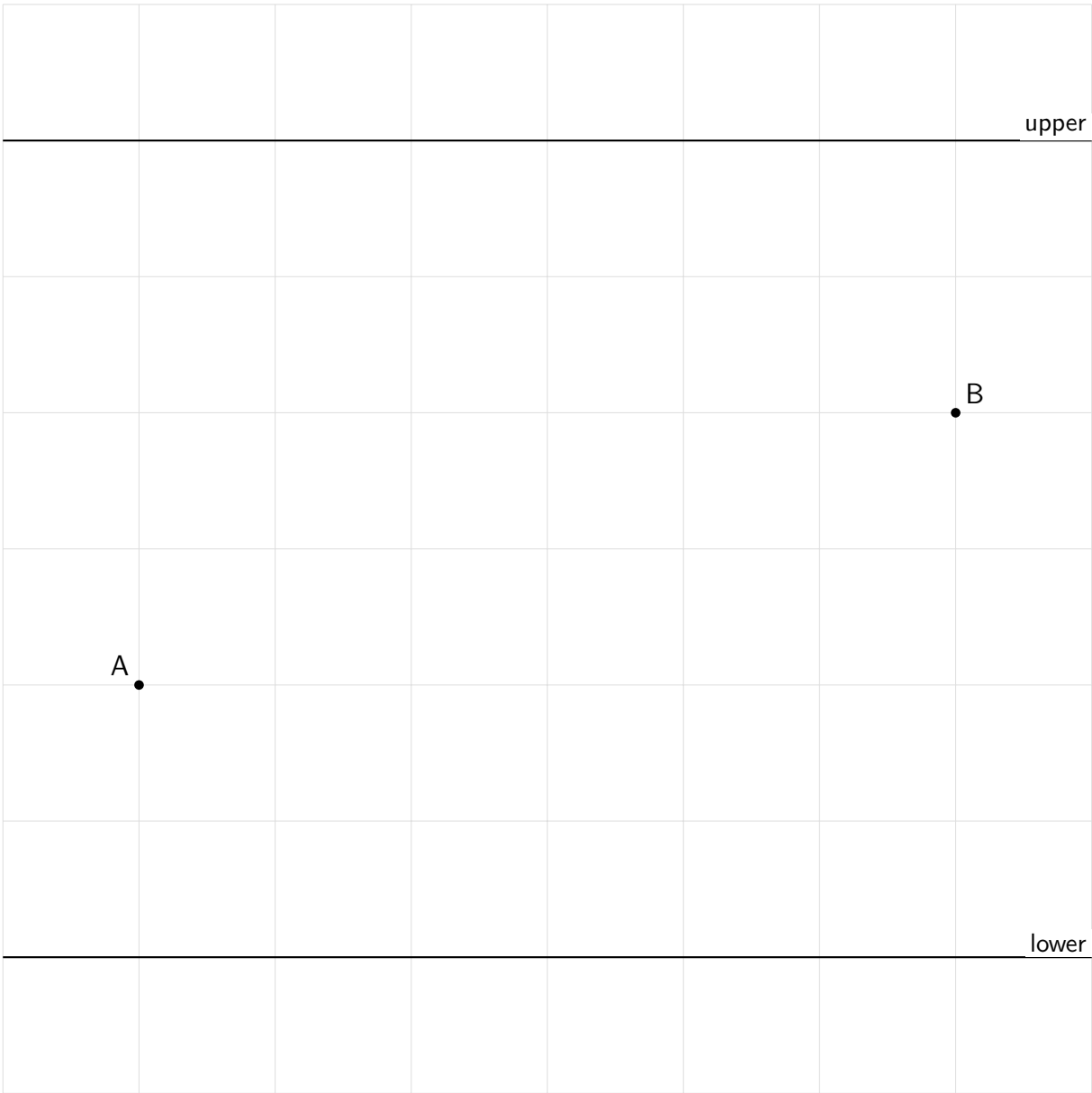


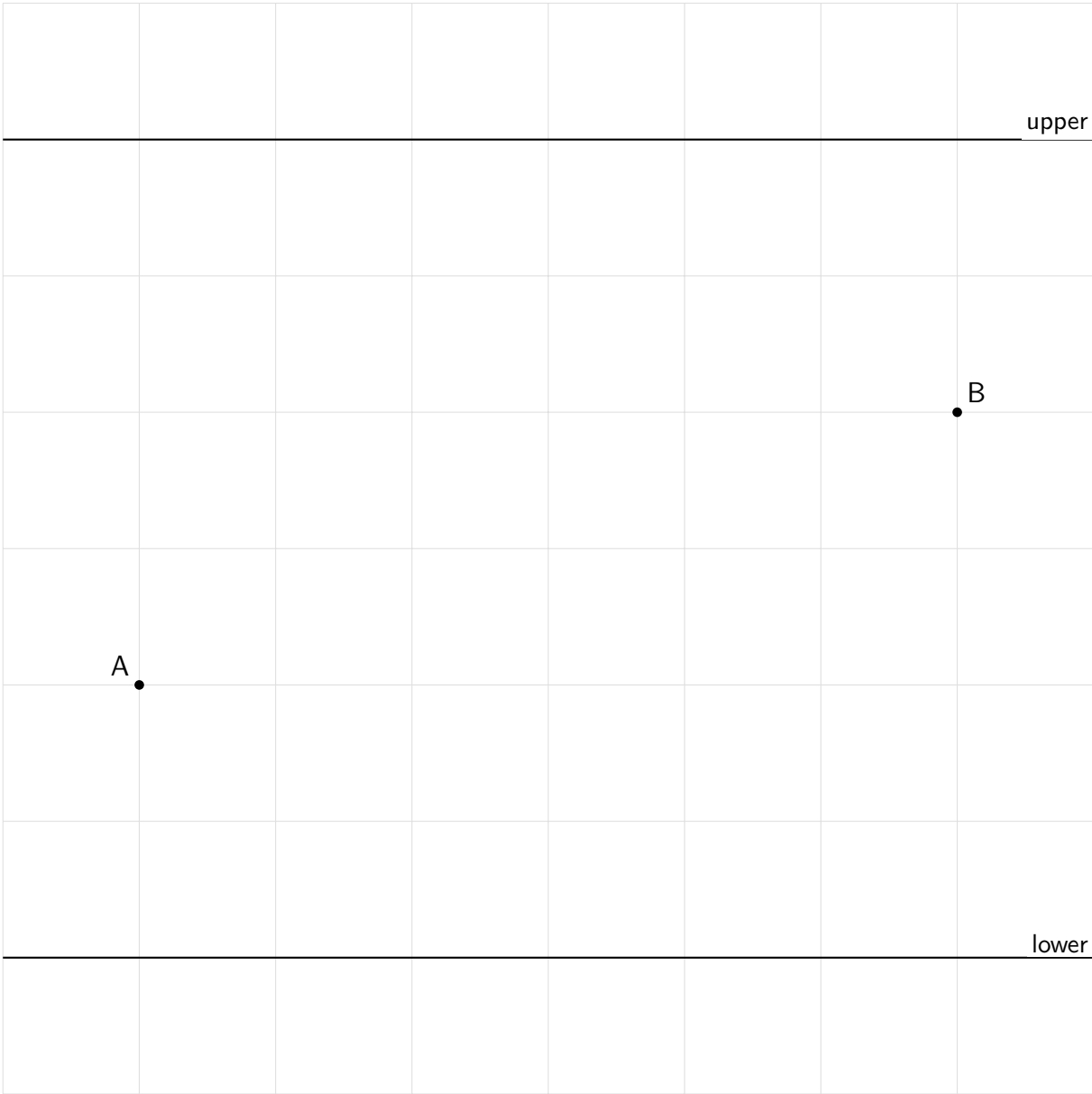
Use straight string pieces or a drawn route. Dot centers are endpoints. For the two-wall boards, stay between the walls, touch them in the stated order, and never travel along a wall.



Problem 1: Find the shortest route from A to B that touches the lower wall first and the upper wall second. Mark both contact points. Explain why a shorter legal route cannot work.

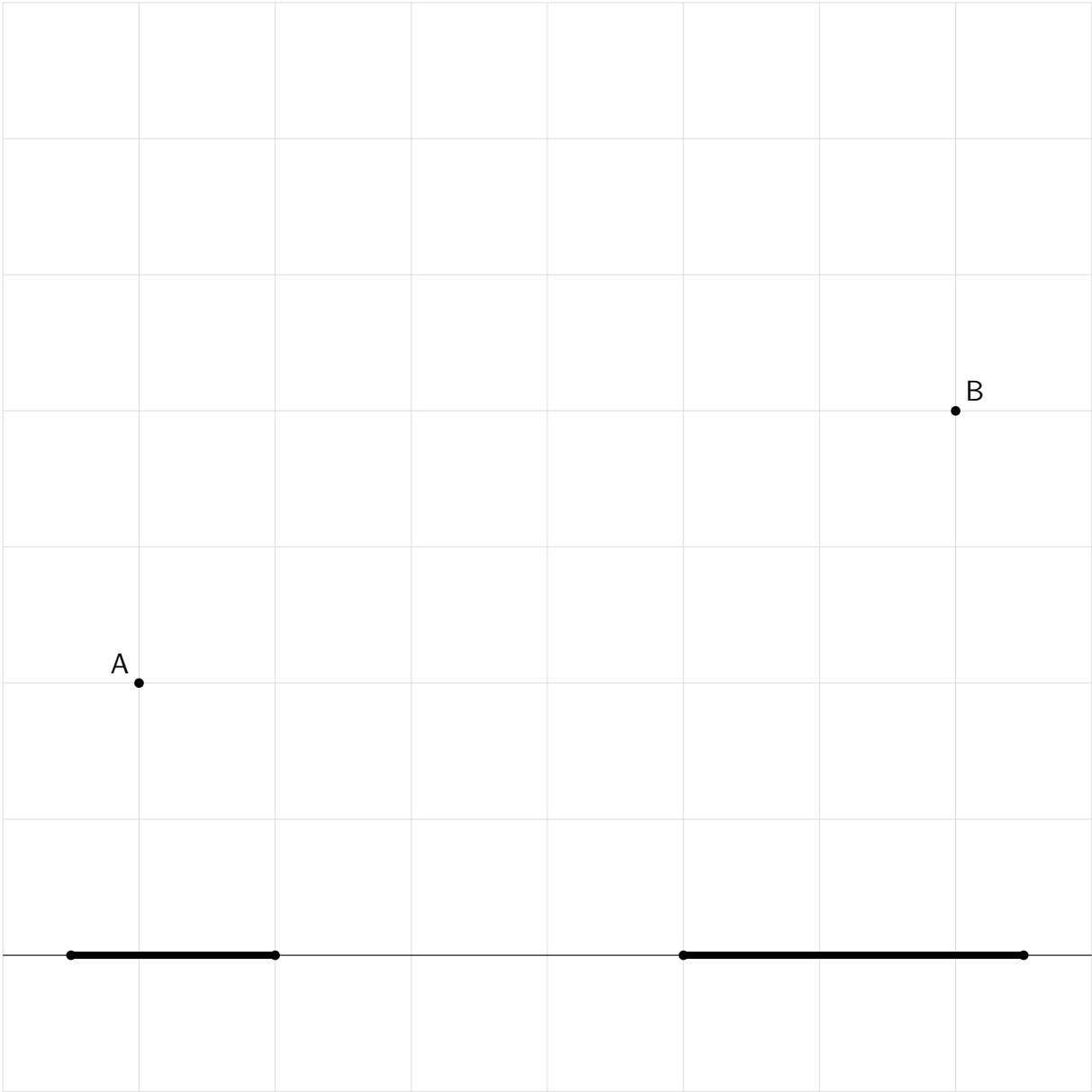


Problem 2: Find the shortest route from A to B that touches the upper wall first and the lower wall second. Which contact order gives a shorter route on this board?



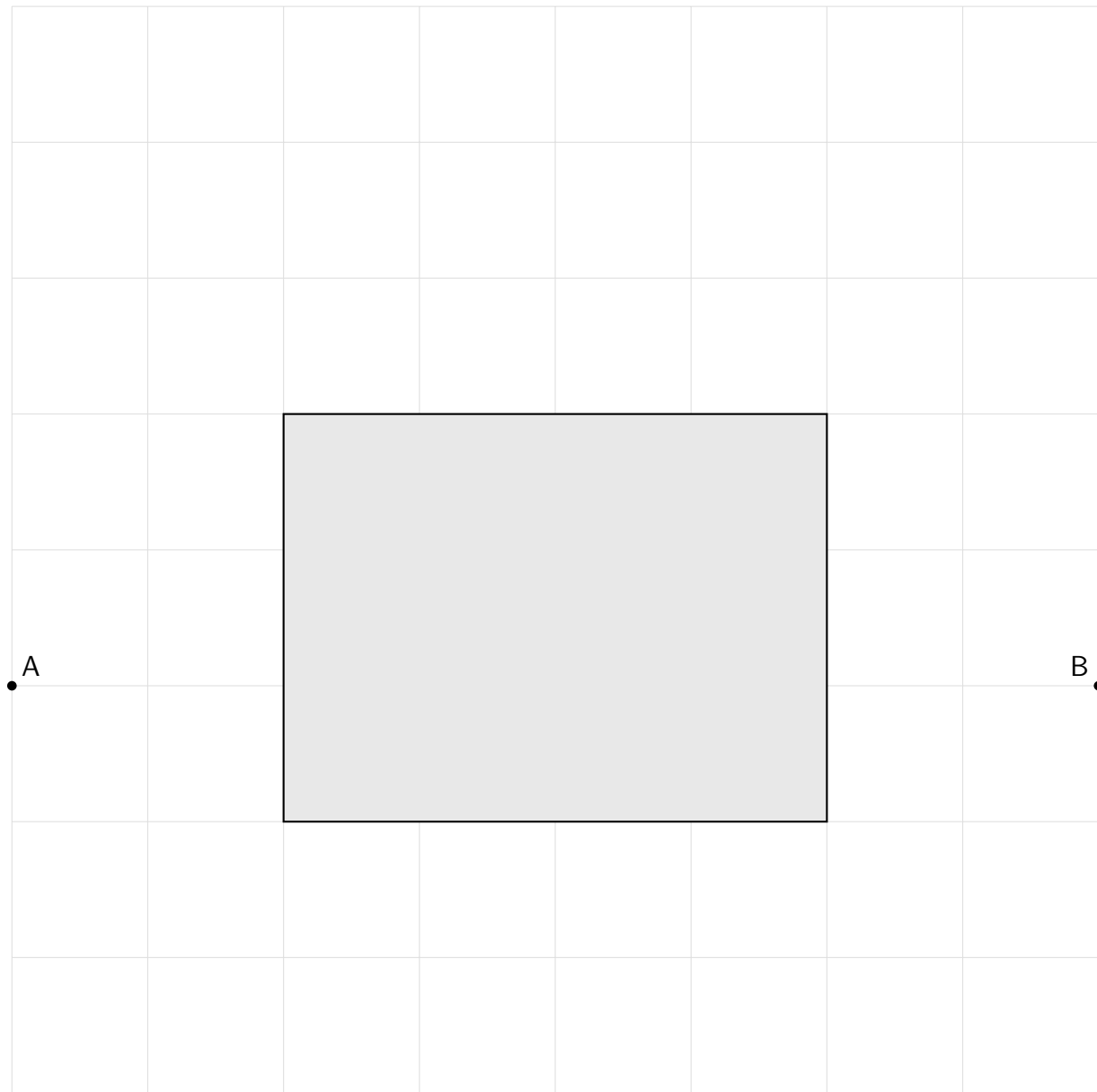
For this board, touch the line exactly once, at a point in one of the two heavy windows. Window endpoints are allowed. Never travel along the line.

Problem 3: Find the shortest allowed route from A to B. Decide which window it uses and where it touches. Explain why every other allowed contact gives a longer route.



The shaded rectangle's interior is forbidden. Touching its edge or traveling along its edge is allowed.

Problem 4: Find the shortest route from A to B around the rectangle. Explain why a route with different bends cannot be shorter.



Problem 5: Find every shortest route from A to B around this rectangle. Its interior is forbidden; its edge is allowed. Can different routes tie for the shortest length?

